

Ideal Line Vortex

$v_\theta = \Gamma/(2\pi r)$ arrow field; circulation and potential flow cartoon.

velocity and radius, explore the concept of circulation as a global property of the flow, and understand how a flow can be irrotational yet still possess a net rotation. The model simplifies real vortices by neglecting viscosity, three-dimensional effects, and core structure, focusing purely on the kinematic consequences of a concentrated vorticity distribution.

Who it's for: Undergraduate students in fluid dynamics or aerodynamics courses studying potential flow theory and vortex dynamics.

KEY TERMS

- Ideal Vortex
- Circulation
- Potential Flow
- Irrotational Flow
- Velocity Field
- Singularity
- Stream Function
- Azimuthal Velocity

How it works

Use with Bernoulli to see low pressure at the core — why bathtubs spin and why wing-tip vortices matter in aviation.

Frequently asked questions

If the fluid is swirling in circles, how can the flow be called 'irrotational'?

In fluid dynamics, 'irrotational' means the local fluid elements do not spin about their own axes (their vorticity is zero). In an ideal line vortex, while the fluid moves in circular paths, each infinitesimal element translates and deforms without net rotation. The 'rotation' we see is a global, orbital motion, not local spinning. This is a key distinction between a vortex and a rotating rigid body.

What does the circulation Γ physically represent, and is it conserved?

Why does the velocity become infinite at the center ($r=0$), and is this realistic?

Where do we see approximations of ideal vortex flow in the real world?

Related simulators

Continue with similar topics in this category — or [all 85 in Classical Mechanics](#).

[View category →](#)

NEW

UNIVERSITY / RESEARCH

Airfoil Streamlines (Joukowski)

▶ Launch Simulator

Potential flow over a Joukowski airfoil: stream-function bands with the Kutta condition fixing circulation.

NEW

UNIVERSITY / RESEARCH

Vortex Ring (Smoke Ring)

▶ Launch Simulator

Self-propelling toroidal vortex shown as a 2D cross-section: counter-rotating Lamb–Oseen pair drifts at...

NEW

UNIVERSITY / RESEARCH

Rocket Propulsion

▶ Launch Simulator

Variable mass: thrust $\dot{m}u$, Tsiolkovsky Δv , vertical launch with gravity.

NEW

KIDS

Rubber Sheet & Ball

▶ Launch Simulator

Sheet height $\propto -\Sigma m/r$; ball rolls along $-\nabla h$ — embedding metaphor, not GR.

NEW

SCHOOL

1D Force Field

▶ Launch Simulator

$U(x)$ presets, $F = -U'$, bead on potential, phase (x,v) and $E(t)$.

NEW

SCHOOL

Ballistic Pendulum

▶ Launch Simulator

Bullet hits block: embedded vs e; $\omega_0 = v/L$, θ_{max} , energy graph.